Survey of KAP of Healthcare providers on antimicrobial resistance in Primary Healthcare of Rwanda.

The knowledge, attitudes and practices of Healthcare providers (doctors, pharmacists, Lab technicians,and nurses) on antimicrobials, antimicrobial resistance and antimicrobial stewardship in Rwanda.

* Indicates required question

Informed consent to participate in a research study

Dear estimated participants,

We kindly request your participation in my research study survey as a prescribing or dispensing or administering health professional.

The aim is to assess the knowledge, attitudes and practices (KAP) of doctors, pharmacists,lab technicians, and nurses on antimicrobials, antimicrobial resistance (AMR) and antimicrobial stewardship (AMS) in Rwanda.

The rise of antimicrobial resistance poses an increasing challenge across both developing and developed nations, impacting various aspects of human health, animal health, the food sector, and agriculture.

Antimicrobials are substances derived from natural, semisynthetic, or synthetic sources that effectively eliminate or suppress the proliferation of microorganisms, all while minimizing harm or damage to the host organism. These include antibacterials, antivirals, antifungals and antiprotozoals.

Antimicrobial resistance arises when microorganisms develop the ability to survive despite the presence of antimicrobial agents that were once effective in treating the infections they cause.

Antimicrobial stewardship is a systematic initiative that encourages the appropriate use of antimicrobials, enhancing outcomes for users, including patients, while mitigating antimicrobial resistance and decrease the transmission of infections caused by multidrug-resistant microorganisms.

The objectives of this study are as follows:

1. To assess the KAP of doctors, pharmacists and nurses on antimicrobials and AMR using a questionnaire survey.
2. To compare and contrast KAP within and between each profession.
3. To train AI-System that would guide on antimicrobials selection,prescription,and consumption.AI-system would advise on interventions,education and training of these professions in terms of antimicrobials, AMR and AMS too.

Your participation in this study involves answering questions about antimicrobials, AMR and AMS. The questions will be open-ended and close-ended and will take approximately 25 min.

Your participation is entirely voluntary.
Please be aware that:

- No identifying information will be stored or disseminated.
- Be assured that answers are anonymous as all surveys will be aggregated first before analysis.
- Confidentiality and anonymity is guaranteed by the electronic survey format.
- Your participation is critical for the success of this research study, and entirely voluntary.
- You are free to decline from participation .
- There are no known risks in participating in this study.
- *This research study has been granted ethical clearance from the College of Medecine and Health Sciences(CMHS)'s Institute Review Board (IRB) of University of Rwanda (Approval No xxxxx).*
- If you wish to proceed and participate after reading this letter, you are giving voluntary informed consent by proceeding with the survey.

xxxx name, Administrator of the College of Medecine and Health Sciences(CMHS)'s Institute Review Board (IRB) , may be contacted on +25(0)xxxxxxxx or xxxxxxxxx@ur.ac.rw.

Thanking you in advance for your participation.

If you have any questions or queries, please feel free to contact the following:

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We thank you in advance for your participation. Yours faithfully, Mukantwari Enatha .

There are 42 questions in this survey.

Demographical information

1. *

What is your gender?

Please choose only one of the following:

Check all that apply.

- [ ] Male
- [ ] Female
- [ ] Prefer not to say

2. **What is your Date of Birth** *

_____

Example: January 7, 2019

3. **Which age group in years do you fall under?** *

Please choose only one of the following:

Check all that apply.

- [ ] 18-24
- [ ] 25-39
- [ ] 40-54
- [ ] 55+

4. **In what year did you graduate?** *

_____

5. **At which University/College/Institute did you obtain your first degree or diploma?**

_____

6. **What qualification did you obtain?** *
For doctors only:

Check all that apply.

- [ ] MBChB (Bachelor of Medicine and Bachelor of Surgery)
- [ ] Master of Medicine
- [ ] Doctorate (PhD in Medicine)
- [ ] Specialist (Fellow of College)
- [ ] Other
- [ ] If other,specify

7. **What qualification did you obtain?** *
For pharmacists only:

Check all that apply.

- [ ] Bachelor of Pharmacy
- [ ] Master of Pharmacy
- [ ] Doctorate (PhD in Pharmacy)
- [ ] Other
- [ ] If other,specify

8. **What qualification did you obtain?** *
For nurses only:

Check all that apply.

- [ ] Diploma in Nursing
- [ ] Bachelor in Nursing
- [ ] Master of Nursing
- [ ] Doctorate or PhD in Nursing
- [ ] Other
- [ ] If other,specify

9. **What is your field of practice?** *

Mark only one oval.

- Public: Primary or community health centre
- Public: District hospital
- Public: Regional hospital
- Public: Tertiary hospital
- Private: Community
- Private: Pharmacy
- Academia
- Other
- I other,specify

10. **Where do you practice?** *

Mark only one oval.

- Urban area
- Peri-urban area
- Rural area

11. **In which province do you practice?** *

Mark only one oval.

- Kigali city
- North
- South
- West
- East

Section 2: Knowledge

12. **Would you consider antimicrobial resistance as a serious problem?** *

Please select your appropriate response for each item:

Check all that apply.

	Strongly agree	Agree	Neutral	Disagree	Strongly Disagree
Globally	☐	☐	☐	☐	☐
In Rwanda	☐	☐	☐	☐	☐
In your hospital or area of practice	☐	☐	☐	☐	☐

13. **How confident are you on your knowledge of antimicrobials, antimicrobial resistance and antimicrobial stewardship?** (Rate yourself on a scale of 1 to 5, with 1 being least confident and 5 being very confident) . *

Check all that apply.

	1	2	3	4	5
Antimicrobials	☐	☐	☐	☐	☐
Antimicrobial resistance	☐	☐	☐	☐	☐
Antimicrobial stewardship	☐	☐	☐	☐	☐

14. **Are antimicrobial drugs effective in treating acute viral infections?** *

Mark only one oval.

- ◯ Yes
- ◯ No

15. **Common colds or flu are caused by:** *

Mark only one oval.

- ◯ Viruses
- ◯ Bacteria

16. **Please select the factors you believe to be the most contributory towards antimicrobial resistance:**

Check all that apply.

- [ ] Overusage of antimicrobials by prescriptions
- [ ] Overusage of antimicrobials without prescriptions
- [ ] Errors in medical prescriptions (dose, duration of use,
- [ ] self-medication & patients not completing prescribed dose,
- [ ] Inadequate hand washing
- [ ] Lack of campaigns on AMR and AMS
- [ ] Lack of new antimicrobial drugs
- [ ] Availability of sub-standards antimicrobial drugs
- [ ] Use of antimicrobials as growth promoters in animals
- [ ] Patient pressure for antimicrobial prescriptions
- [ ] Limited time to spend with patients due to having more patients to consults.
- [ ] over-the-counter access to antimicrobials
- [ ] weak supply chain management and proliferation of substandard and falsified antimicrobials
- [ ] lack of laboratory capacity to identify the organism and its antimicrobial susceptibility
- [ ] lack of knowledge or up-to-date information on prescription of antimicrobials
- [ ] lack of treatment guidelines
- [ ] unreliable or absent surveillance data on AMR and antimicrobial usage
- [ ] unregulated over-the-counter use and poor antimicrobial stewardship (AMS
- [ ] Limited quality teaching resources to address education in AMR
- [ ] Private Pharmacy owners provides antimicrobial drug without prescription to make more money.

17. **Which of the following sources do you refer to for information on appropriate usage of antimicrobials?** Please select your appropriate choice and rate on a scale of 1 to 5 (1 = lowest preference and 5 = highest preference):

Mark only one oval per row.

	1	2	3	4	5
Information from senior experienced colleagues	◯	◯	◯	◯	◯
Rwanda Medicines Formulary	◯	◯	◯	◯	◯
The Merck Manual	◯	◯	◯	◯	◯
Medical Journals or Wbsites	◯	◯	◯	◯	◯
Standard guidelines	◯	◯	◯	◯	◯
International guidelines(e.g.Professional society guidelines)	◯	◯	◯	◯	◯

18. **Please assess your levels of confidence in the following areas when prescribing antimicrobials?** (1 = very confident, 2 = confident, 3 = unsure, 4 = unconfident, 5 = very unconfident and 6 = not applicable) *

Mark only one oval per row.

	1	2	3	4	5
Making an accurate diagnosis of the infection	◯	◯	◯	◯	◯
Decision not to prescribe antimicrobial when a patient has a fever with no seriuos criteria and you are not sure of the diagnosis	◯	◯	◯	◯	◯
Selecting the correct antimicrobial	◯	◯	◯	◯	◯
Selecting the correct dosage for the antimicrobial	◯	◯	◯	◯	◯
Selecting the correct intervl for antimicrobial	◯	◯	◯	◯	◯
Selecting the correct duration for antimicrobial	◯	◯	◯	◯	◯

Using

Using combination therapy if necessary					
Interpreting microbiological laboratory results					
When to stop or change the antimicrobial therapy					
Selecting between an intravenous or oral antimicrobials					

19. **Do you think there are risks associated with irrational use of antimicrobials?** *

Mark only one oval.

- Yes
- No
- I do not know
- If you answered Yes,Please list the risks.

20. **Do you think restricting antimicrobial usage is necessary to reduce antimicrobial resistance?** *

Mark only one oval.

- Yes
- No
- Maybe

21. **Have you attended any workshops or had training on antimicrobials and antimicrobial stewardship?** *

Mark only one oval.

- Yes
- No

22. **Would you like more education and training on antimicrobial use, antimicrobial resistance and antimicrobial stewardship?** *

Mark only one oval.

- Yes
- No

23. **Please list the three most resistant bacteria you are aware o**f (with 1 being the most resistant and 3 being the least resistant): *

24. **What percentage of clinical antimicrobial use is unnecessary or inappropriate in Rwanda?** *

Mark only one oval.

- Between 1% and 20%
- Between 21% and 40%
- Between 41% and 60%
- etween 61% and 80%
- Between 81% and 100%

25. **Have you heard of the term Defined Daily Dose (DDD) as used by the World Health Organization?** *

Mark only one oval.

- Yes
- No

26. **What is the DDD value assigned by WHO for ceftriaxone in adult patients?** * Dropdown

Mark only one oval.

- 500 mg
- 1 g
- 2 g
- I don't know

27. **Do you know the purpose of the WHO DDD metric?** * Dropdown

Mark only one oval.

- For patient dosing
- For policy and surveillance of antibiotic use
- For pharmacy stock control
- I don't know

28. **Do you know the national guidelines for prescribing antibiotics for common infections (e.g., UTI, pneumonia)?**

Mark only one oval.

- Yes
- No

29. **What laboratory test is used to determine if a bacterial infection is resistant to antibiotics?**

Section 3: Attitudes

30. **Antimicrobials are safe drugs that can be commonly prescribed?** *

Mark only one oval.

- Yes
- No

31. **Prescribing antimicrobials to a patient who does not really need them may ultimately have a negative impact on their health**. Please choose one option from the list *

Mark only one oval.

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

32. **In your opinion, do you think the problem of antimicrobial resistance is getting better or worse?** *

Mark only one oval.

- Better
- Worse

33. **What do you think are the important strategies to combat antimicrobial resistance?** *

Select the most appropriate choices:

Check all that apply.

- [ ] Educational campaigns
- [ ] Use of therapeutic guidelines
- [ ] Vaccination /AMR campaigns
- [ ] Improved infection control
- [ ] Reducing antimicrobial use in agriculture and animals
- [ ] Better control on antimicrobial sales
- [ ] Other
- [ ] If other, specify:

34. **Do you believe a patient skipping or missing one or two doses of antimicrobials contributes to the development of antimicrobial resistance?** *

Mark only one oval.

- ◯ Yes
- ◯ No

35. **Do you believe that new classes of antimicrobials will be developed in the next 5–10 years?** *

Mark only one oval.

- ◯ Yes
- ◯ No

36. **Do you believe that you may personally be contributing towards antimicrobial resistance?** *

Mark only one oval.

- Yes
- No
- Maybe

37. **Do you believe knowing the DDD of antibiotics can help improve prescribing practices?** * Dropdown

Mark only one oval.

- Strongly agree
- Agree
- Disagree
- Strongly disagree

38. **Should national guidelines include WHO DDD values for training and awareness?** * Dropdown

Mark only one oval.

- Yes
- No
- May be

39. **Do you agree or disagree: "It is acceptable to prescribe antibiotics without laboratory confirmation if the patient insists"?** *

Mark only one oval.

- Strongly disagree
- Disagree
- Neutral
- Agree
- Strongly agree
- In some situations or conditions

40. **How important is AST in guiding your antibiotic prescription? (Very important → Not important)** Dropdown

Mark only one oval.

- Very imoortant
- Important
- Not important

Section 4: Practice

41. **For pharmacists only: In addition to your Bachelor of Pharmacy degree, do you possess permit allowing you to prescribe for primary care drug treatment?** *

Mark only one oval.

- Yes
- No

42. **For nurses only: Do you possess a concession permit to prescribe medicines to patients.** *

Mark only one oval.

- Yes
- No

43. **Do you prescribe antimicrobial drugs to your patients?** *

Mark only one oval.

- Yes
- No
- Not applicable

44. **Please select the average number of patients to whom you prescribe antimicrobials per day?** *

Mark only one oval.

- Not Applicable
- Less than 5
- Between 6 and 10
- Between 11 and 20
- Between 21 and 30
- Between 31 and 40
- Greater than 40

45. **Do you prescribe antimicrobials empirically (based on observation and experience)?** *

Mark only one oval.

- Yes
- No
- Not applicable

46. **Before initiating antimicrobial therapy, do you send samples for microbiology testing to inform the need for and/or choice of antimicrobial therapy** *

Mark only one oval.

- Yes
- No
- Not applicable

47. **Do you possess the latest Rwanda Standard Treatment Guidelines?** *

Mark only one oval.

- Yes
- No

48. **How often do you use the antimicrobial treatment guidelines when deciding upon antimicrobials for a patient?** *

Mark only one oval.

- Always
- Sometimes
- Never
- Not applicable

49. **Which three antimicrobials do you prescribe the most (if applicable)** *

_____

50. **Have you changed your prescribing behaviour in light of antimicrobial resistance in the past 5 years** *

Mark only one oval.

- Yes
- No
- Not applicable
- If Yes,how?

51. **What advice do you impart to patients on antimicrobial usage?** *

Mark only one oval.

- The patient must complete the full course of antimicrobials for the recommended duration
- The patient must take the antimicrobials at the correct time intervals
- The patient should never share antimicrobial treatment with anyone
- The patient must not stop treatment even if they are feeling better after a few doses
- The patient must save the remaining antimicrobials for the next time they are unwell.
- The patient must dispose of remaining antimicrobials appropriately

52. **When prescribing antibiotics like ceftriaxone, what factors do you consider most? (Tick all that apply)** *

Check all that apply.

- [ ] Clinical presentation
- [ ] Local guidelines
- [ ] AST results
- [ ] DDD benchmarks

53. **Do you currently use DDD-based data or reports in your decision-making?** Dropdown

Mark only one oval.

- Yes
- No
- Not Sure

54. **In the past month, how often have you prescribed antibiotics without conducting or receiving AST results?** Dropdown

Mark only one oval.

- Always
- Sometimes
- Never

55. If you answered that you sometime or never conduct or wait for AST results before prescribing antibiotics,Please tell us what were the main three raisons? *

56. **Have you prescribed antibiotics for conditions you suspected to be viral in origin? (Yes/No)** *

Mark only one oval.

- Yes
- No

57. **How often do you follow national or WHO guidelines when prescribing antimicrobials?**

Mark only one oval.

- Always
- Often
- Sometimes
- Rarely
- Never

58. **Do you routinely document the diagnosis, antibiotic indication, and dosage in the patient file?** *

Mark only one oval.

- Yes
- No
- Sometimes

59. **Which antibiotics do you most frequently prescribe as first-line treatment?** *

Antimicrobial treatment

60. **Please provide the required information on which condition from the list you request for laboratory culture.** *

For Doctors only:

Mark only one oval per row.

	Always-A	On treatment failure – B
Acute pharyngitis		
Nasopharyngitis		
Acute otitis media		
Chronic otitis media		
Acute sinusitis		
Bronchitis		
Pneumonia		
Cystitis		
Pyelonephritis		

61. **What are your first choice of antimicrobials and Adjunct or other treatment for Acute pharyngitis ?** *

For Doctors only.

62. **What are your first choice of antimicrobials and Adjunct or other treatment for Nasopharyngitis ?** *

For Doctors only.

63. **What are your first choice of antimicrobials and Adjunct or other treatment for Acute otitis media ?** *

For Doctors only.

64. **What are your first choice of antimicrobials and Adjunct or other treatment for Chronic otitis media ?** *

For Doctors only.

65. **What are your first choice of antimicrobials and Adjunct or other treatment for Acute sinusitis ?** *

For Doctors only.

66. **What are your first choice of antimicrobials and Adjunct or other treatment for Bronchitis ?** *

For Doctors only.

67. **What are your first choice of antimicrobials and Adjunct or other treatment for Pneumonia ?** *

For Doctors only.

68. **What are your first choice of antimicrobials and Adjunct or other treatment for Cystitis ?** *

For Doctors only.

69. **What are your first choice of antimicrobials and Adjunct or other treatment for Pyelonephritis ?** *

For Doctors only.

70. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Acute diarrhoea in adults.

71. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Chronic diarrhoea in adults .

72. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Acne vulgaris of skin .

73. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for urinary tract infections of uncomplicated cystitis in adults.

74. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for urinary tract infections of complicated cystitis in adults.

75. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for urinary tract infections of complicated cystitis in pregnant women.

76. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Respiratory conditions of Acute bronchitis in adults and adolescents .

77. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Respiratory conditions of Pneumonia uncomplicated (excludes paediatric and over 65 years) .

78. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for eye infections of Bacterial eye infections, conjunctivitis (excludes newborn) .

79. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Ear, nose and throat conditions of Otitis externa .

80. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Ear, nose and throat conditions of Otitis media acute .

81. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Ear, nose and throat conditions of Sinusitis, acute, bacterial .

82. **For pharmacists only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Ear, nose and throat conditions of Tonsillitis and pharyngitis .

83. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Severe necrosing gingivitis .

84. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Tick bite fever .

85. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Acute meningitis .

86. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Conjunctivitis.

87. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Acute otitis media .

88. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Acute sinusitis .

89. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Bacterial tonsillitis .

90. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Acute bronchitis .

91. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Boil abscess .

92. **For nurses only:** *

Please provide which antimicrobial drugs you are authorised to prescribe for the following conditions (if applicable) for Urinary tract infection, uncomplicated .

Open-Ended Questions for everyone

93. *

Can you describe a recent situation where you had to decide whether or not to prescribe antibiotics? What factors influenced your decision?

94. *

What challenges do you face when trying to follow antibiotic guidelines in your facility?

95. *

How do patients typically react when you decide not to prescribe antibiotics?

96. *

What resources or support would help you feel more confident in antimicrobial prescribing?

97. *

How would you explain antimicrobial resistance to a patient in simple terms?

98. *

What would a successful antimicrobial stewardship program look like in your context?

99. *

Can you describe the gaps between national AMR policies and implementation at health facilities?

100. *

What kind of behavior change is needed at the community or provider level to reduce AMR?

101. *

What do you think are the biggest system-level challenges to controlling AMR in Rwanda?

102. *

How do you see the role of technology, like AI, in improving antimicrobial use?

103. *

“A mother insists on antibiotics for her child’s viral fever. How would you respond?”

Mark only one oval.

- Prescribe to satisfy her
- explain AMR risks
- suggest alternatives.

104.

“**A farmer uses leftover animal antibiotics for his own infection. How would you advise him?”**

_____

Rwanda context-specific questions

105. **About Informal Drug Vendors and Access:**

How often do patients in your community obtain antibiotics without prescriptions (e.g., from local pharmacy, health facility, or animal care facility)?

_____

106. **About Informal Drug Vendors and Access:** *

Do you collaborate with community health workers (Binômes) to address inappropriate antibiotic use? If not, why?

_____

107. **About Informal Drug Vendors and Access:** *

Are informal vendors in your area trained on AMR risks or regulated by local authorities?

_____

108. *

Mobile Health (mHealth) and Infrastructure:

Do you use e-Ubuzima or other digital tools report AMR cases or access treatment guidelines?

_____

109. **Mobile Health (mHealth) and Infrastructure:** *

How does geographic remoteness (e.g., lack of roads, mountainous terrain) impact your ability to manage infections appropriately?

_____

110. *

For Traditional Medicine Beliefs:

Do you believe traditional remedies (e.g., umuravumba, igikakarubamba) can cure infections without antibiotics?

_____

111. **For Traditional Medicine Beliefs:** *

Have you ever suggested to your patients traditional medicine instead of antibiotics?

_____

112. **For Economic and Insurance Factors:** *

Do you think that the cost of visiting a clinic (even with Mutuelle de Santé insurance) can influence your patients to make decision to buy antibiotics informally?

_____

113. **For Cultural and Social Nuances:** *

Do you think antibiotics are seen as “strong medicine” (ubuvuzi bukomeye) in your community?

_____

114. **For Cultural and Social Nuances:** *

Are there local terms or phrases that describe AMR (e.g., “ubuvuzi butazigera bukora” – medicine that no longer works)?

_____

115. **For Cultural and Social Nuances:** *

How do religious or community leaders influence health-seeking behaviors (e.g., advising prayer over antibiotics)?

_____

116. *

For Integration with Rwanda's Healthcare System:

How does performance-based financing (e.g., healthcare worker incentives) affect antibiotic prescribing habits?

117. **For Integration with Rwanda's Healthcare System:** *

Are maternal and child health programs (e.g., PMTCT) integrated with AMR stewardship efforts?

118. **For Integration with Rwanda's Healthcare System:** *

Do community health insurance schemes (Mutuelle de Santé) cover diagnostic tests to reduce unnecessary antibiotic use?

Aknowledgement and Submission

Thank you for your valuable time in participating in this study. Your response will be invaluable towards optimising rational antimicrobial use
and AMS.

Thank you for completing this survey.

You can submit your survey now.

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